

DATA REPRESENTATION PRACTICE SHEET

NIMCET TARGET PAPER • ADVANCED LEVEL (HARD)

ResourceOS

1. In IEEE 754 single-precision floating-point format, what is the exact decimal value represented by the 32-bit hex word 0xC0A00000?

- (A) -2.5 (B) -5.0
(C) -1.25 (D) -10.0

2. A machine uses an 8-bit 2's complement format. What are the minimum and maximum decimal integers that can be represented, and what is the 2's complement representation of the smallest negative number?

- (A) -128 to +127; 10000000 (B) -127 to +128; 11111111
(C) -128 to +127; 11111111 (D) -256 to +255; 10000000

3. Convert the repeating decimal fraction 0.375_{10} to base-5 representation.

- (A) $0.1414..._5$ (B) $0.1421..._5$
(C) $0.1420..._5$ (D) $0.1341..._5$

4. Which of the following operations in 8-bit 2's complement arithmetic causes an internal overflow condition?

- (A) $01100100 + 00011001$ (B) $01000000 + 01000001$
(C) $11000000 + 11010000$ (D) $11111111 + 00000001$

5. In a Hamming (7, 4) code with even parity, if the received code word is 1011011 (bit positions 1 to 7 from left to right, parity bits at 1, 2, 4), which bit position contains an error?

- (A) Bit 3 (B) Bit 5
(C) Bit 6 (D) No error detected

6. What is the 16's complement of the hexadecimal integer 2A3F?

- (A) D5C0 (B) D5C1
(C) D5BF (D) C5C1

7. What is the difference between the largest and smallest positive normalized floating-point numbers in IEEE 754 single precision (approximate relative order)?

- (A) $\sim 10^{38}$ and $\sim 10^{-38}$ (B) $\sim 10^{308}$ and $\sim 10^{-308}$
(C) $\sim 10^{16}$ and $\sim 10^{-16}$ (D) $\sim 10^7$ and $\sim 10^{-7}$

8. Convert the Gray code 1101011 into its original binary equivalent.

- (A) 1001101 (B) 1010010
(C) 1001110 (D) 1110011

9. How many distinct zero representations exist in 8-bit Signed Magnitude, 1's Complement, and 2's Complement systems, respectively?

- (A) 2, 2, 1 (B) 1, 2, 1
(C) 2, 1, 1 (D) 1, 1, 1

10. In IEEE 754 single-precision standard, a subnormal (denormalized) number is indicated when the exponent bits are:

- (A) All 0s and mantissa is non-zero
- (B) All 1s and mantissa is zero
- (C) All 1s and mantissa is non-zero
- (D) All 0s and mantissa is zero

11. Calculate $(734)_8 - (475)_8$ directly in octal arithmetic.

- (A) $(237)_8$
- (B) $(247)_8$
- (C) $(257)_8$
- (D) $(337)_8$

12. What is the machine epsilon (ϵ) for IEEE 754 single-precision floating-point arithmetic?

- (A) 2^{-23} (approx. 1.19×10^{-7})
- (B) 2^{-52} (approx. 2.22×10^{-16})
- (C) 2^{-127}
- (D) 2^{-24}

13. Perform 2's complement subtraction of -18 from -12 using an 8-bit allocation. What is the resulting 8-bit binary pattern?

- (A) 00000110
- (B) 11111010
- (C) 00000100
- (D) 11111100

14. What decimal fraction corresponds exactly to the binary fractional pattern $0.0001100110011\dots_2$ (with '0011' repeating infinitely)?

- (A) $1/10$
- (B) $1/12$
- (C) $1/15$
- (D) $3/20$

15. What is the 7-bit 2's complement representation of -64?

- (A) 1000000
- (B) 1111111
- (C) Cannot be represented in 7 bits
- (D) 1000001

16. If r is the radix of a number system, what is the formula for the $(r - 1)$'s complement of an n -digit integer N ?

- (A) $(r^n - 1) - N$
- (B) $r^n - N$
- (C) $r^{n-1} - N$
- (D) $(r^n + 1) - N$

17. Convert decimal -0.75 to IEEE 754 32-bit single-precision floating-point representation in Hexadecimal.

- (A) 0xBF400000
- (B) 0x3F400000
- (C) 0xBF800000
- (D) 0xC0000000

18. What is the minimum number of bits required to uniquely represent all signed integers in the range -250 to +250 in 2's complement form?

- (A) 8
- (B) 9
- (C) 10
- (D) 11

19. Which property distinguishes a Quiet NaN (qNaN) from a Signaling NaN (sNaN) in IEEE 754 floating-point standard?

- (A) The MSB of the mantissa field is 1 for qNaN and 0 for sNaN. (B) Exponent bits are all 0s for qNaN and all 1s for sNaN.
(C) Sign bit is 0 for qNaN and 1 for sNaN. (D) Mantissa is zero for qNaN and non-zero for sNaN.

20. Compute the BCD addition of decimal 68 and 57. What is the final 12-bit corrected BCD result?

- (A) 0001 0010 0101 (B) 0001 0010 1100
(C) 0000 0111 1101 (D) 0001 0011 0101

21. In arithmetic right shift of an 8-bit signed binary integer 10110100 by 2 positions, what is the resulting bit pattern?

- (A) 11101101 (B) 00101101
(C) 11011010 (D) 10110111

22. Convert the base-7 number $(56.4)_7$ to base-10 decimal.

- (A) 41.5714... (B) 41.2857...
(C) 39.5714... (D) 41.8571...

23. What is the main advantage of Gray code over standard binary sequence in rotary optical encoders?

- (A) Only one bit changes state between adjacent consecutive values, eliminating spurious transition errors. (B) It allows direct addition without correction cycles.
(C) It supports self-complementing arithmetic. (D) It uses fewer bits to encode the same decimal range.

24. In a 12-bit fixed-point binary representation with 1 sign bit, 5 integer bits, and 6 fractional bits, what is the largest positive representable value?

- (A) 31.984375 (B) 31.500000
(C) 63.984375 (D) 15.984375

25. In 1's complement arithmetic, adding binary 11110100 (-11) to 00001101 (+13) yields an end-around carry. What is the corrected final sum?

- (A) 00000010 (+2) (B) 00000001 (+1)
(C) 00000011 (+3) (D) 11111110 (-1)

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(B)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(C)	13	(A)	18	(B)	23	(A)
4	(B)	9	(A)	14	(C)	19	(A)	24	(A)
5	(B)	10	(A)	15	(A)	20	(A)	25	(A)